S/N	Item	Description	Qty
1	SG90 9G Micro Servo	9 g micro servo motor used for pan/tilt positioning.	2
2	Raspberry Pi 2011	Raspberry Pi Model B used as the higher-level computer/host.	1
3	Arduino Mega	Arduino Mega microcontroller used as the main hardware I/O controller.	1
4	Raspberry Pi camera	Raspberry Pi camera module used for visual sensing and imaging.	1
5	SparkFun QRE1113 Digital Line Sensor Breakout	QRE1113 infrared reflective line sensor breakout used for line/reflective-surface detection.	3
6	Sharp GP2Y0A02YK IR distance sensor	Sharp GP2Y0A02YK infrared distance sensor used for non-contact distance measurement.	1
7	9 DOF sensor stick LMS9DS1	LMS9DS1 9-axis inertial sensor containing accelerometer, gyroscope and magnetometer.	1
8	DC-DC voltage converter module	DC-DC voltage converter used to regulate battery voltage for control electronics.	1
9	SD HC card	SDHC memory card used for data/storage or Raspberry Pi operating-system storage.	1
10	OPB706B IR diode / phototransistor	OPB706B infrared diode and phototransistor sensor used for optical detection.	3
11	Power switch	Power switch used to turn the robot electrical supply on and off.	1
12	802.11n USB Wi-Fi adapter	802.11n USB Wi-Fi adapter providing wireless network connectivity.	1
13	SparkFun logic level converter	Logic-level converter used to translate digital signals between 5 V and 3.3 V logic.	1
14	Custom PCB	Custom interface PCB used to interconnect the Arduino and peripheral devices.	1
15	L293DNE	L293DNE quadruple half-H motor driver IC used to drive DC motors.	1
16	Caster wheel	Caster wheel providing passive mechanical support for the mobile robot.	1
17	5-AA battery pack	5-AA battery pack providing electrical power to the robot.	1
18	Geared DC motor	Geared DC motors providing drive torque to the robot wheels.	2
19	Motor encoder SEN0038	SEN0038 motor encoder used to measure wheel/motor rotation.	2
20	H-bridge motor driver	H-bridge motor driver used to control motor direction and speed.	1
21	Robot chassis	Robot chassis providing the mechanical structure for mounting components.	1
22	2-inch Robot tyres	2-inch robot tyres providing traction and motion.	2
23	Wires	Electrical wires used to interconnect power, sensors, motors and control electronics.	